I SEMESTER

ADVANCED CALCULUS

Course Code: MA11 **Credits: 3:1:0**

Pre-requisites: - **Contact Hours: 42L + 14T**

Course Coordinators: Dr. S.H.C.V. Subba Bhatta & Mr. B. Azghar Pasha

Unit-I

Differential Calculus-I: Polar curves, Angle between the radius vector and the tangent, Angle between the curves, Length of perpendicular from pole to the tangent, Pedal equations.
Derivative of arc length & Radius of Curvature in Cartesian, polar & parametric forms (All without proof)
Partial Differentiation: Partial derivatives, Total differential coefficient, Differentiation of composite and implicit functions, Euler's theorem. Jacobians and properties.

- Pedagogy/Course delivery tools: ➢ Chalk and talk
- Links: ➢ https://nptel.ac.in/courses/111/105/111105121/
- Impartus recording: ➢ https://a.impartus.com/ilc/#/course/107625/1030

Unit-II

Integral Calculus - I: Reduction formulae for $sin^n x$, $cos^n x$, $sin^m x\ cos^n x$, Evaluation of these integrals with standard limits, Tracing of curves (both Cartesian and polar)-Illustration of the same using online tools or open source software. Beta and Gamma functions-Introduction, properties (without proof).
Application of integration – length of arc of a curve, plane areas, volumes and surface area of revolution

- Pedagogy/Course delivery tools: ➢ Chalk and talk, PowerPoint Presentation,Videos
- Online tool & Open source software: ➢ Use of Online tools such as **DESMOS** or open-source software such as **Scilab** to demonstrate the characteristics of graphs
- Links ➢ https://nptel.ac.in/courses/111/104/111104144/
- Impartus recording: ➢ https://a.impartus.com/ilc/#/course/107625/1030

Unit-III

Vector differentiation: Vector functions of a single variable, Derivative of a vector function, Geometrical interpretation, Velocity and acceleration.

Scalar and vector fields, Gradient of a scalar field, Directional derivative, Divergence of a vector field, Solenoidal vector, Curl of a vector field, Irrotational vector, Laplacian operator, Vector identities.

• Pedagogy/Course delivery tools:	➢ Chalk and talk
• Links:	➢ https://nptel.ac.in/courses/111/105/111105134/
• Impartus recording:	➢ https://a.impartus.com/ilc/#/course/107625/1030

Unit-IV

Integral Calculus - II: Multiple integrals- evaluation of double and triple integrals, Change of order of integration, Change of variables. Applications of double and triple integrals to find areas and volumes. Evaluation of multiple integrals using online tools or open source software.

• Pedagogy/Course delivery tools:	➢ Chalk and talk, PowerPoint Presentation,Videos
• Online tool & Open source software:	➢ Demonstration of evaluation of multiple integrals using open source software
• Links:	➢ https://nptel.ac.in/courses/111/105/111105121/
• Impartus recording:	➢ https://a.impartus.com/ilc/#/course/107625/1030

Unit-V

Vector integration: Line integrals, surface integrals and volume integrals. Green's theorem (with proof) and its applications, Stokes' theorem and Gauss divergence theorem (without proofs) and its applications.

• Pedagogy/Course delivery tools:	➢ Chalk and talk, Videos
• Links:	➢ https://nptel.ac.in/courses/111/105/111105134/
• Impartus recording:	➢ https://a.impartus.com/ilc/#/course/107625/1030

Text Books:

1. **George B. Thomas, Maurice D. Weir, Joel R. Hass** - Thomas' Calculus, Pearson, 13th edition, 2014.
2. **B.S. Grewal** – Higher Engineering Mathematics, Khanna Publishers, 44th edition, 2017.

Reference Books:

1. **Erwin Kreyszig** – Advanced Engineering Mathematics, Wiley publication, 10th edition, 2015.

2. **Srimanta Pal & Subodh C Bhunia** - Engineering Mathematics, Oxford University Press, 3^{rd} Reprint, 2016.
3. **B. V. Ramana** - Higher Engineering Mathematics, Tata McGraw-Hill, 11^{th} edition, 2010.

Course Outcomes (COs):

At the end of the course the student will be able to

1. Solve problems related to Polar curves, Radius of curvature and Jacobians. (PO-1, 2)
2. Trace a curve using its guiding properties and use integration to find its perimeter, surface area and volume. (PO-1, 2)
3. Apply vector differentiation to identify solenoidal and irrotational vectors. (PO-1, 2)
4. Evaluate multiple integrals and use them to find areas and volumes. (PO-1, 2)
5. Exhibit the interdependence of line, surface and volume integrals using integral theorems. (PO-1, 2)

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	**Marks**	**Course outcomes addressed**
Internal Test-I	30	CO1,CO2,CO3
Internal Test-II	30	CO3,CO4,CO5
Average of the two internal tests shall be taken for 30marks.		
Other components	**Marks**	**Course outcomes addressed**
Quiz	10	CO1,CO2,CO3
Assignment	10	CO3,CO4,CO5
Semester End Examination:	100	CO1,CO2,CO3,CO4,CO5